

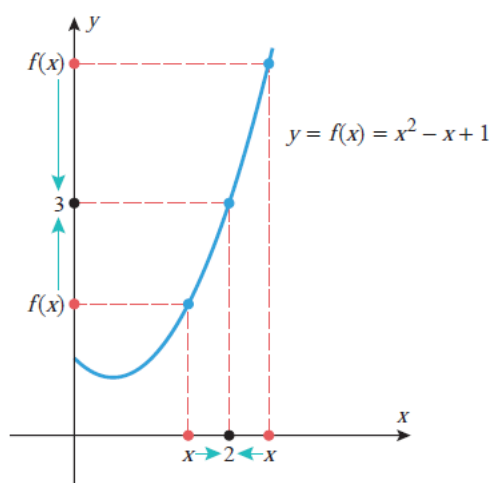
Limits:

The most basic use of limits is to describe how a function behaves as the independent variable approaches a given value. For example, let us examine the behavior of the function

$$f(x) = x^2 - x + 1$$

for x -values closer and closer to 2. It is evident from the graph and table in Figure 1.1.8 that the values of $f(x)$ get closer and closer to 3 as values of x are selected closer and closer to 2 on either the left or the right side of 2. We describe this by saying that the “limit of $x^2 - x + 1$ is 3 as x approaches 2 from either side,” and we write

$$\lim_{x \rightarrow 2} (x^2 - x + 1) = 3 \quad (5)$$



1.1.1 LIMITS (AN INFORMAL VIEW) If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a), then we write

$$\lim_{x \rightarrow a} f(x) = L \quad (6)$$

which is read “the limit of $f(x)$ as x approaches a is L ” or “ $f(x)$ approaches L as x approaches a .” The expression in (6) can also be written as

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a \quad (7)$$

ONE-SIDED LIMITS

The limit in (6) is called a *two-sided limit* because it requires the values of $f(x)$ to get closer and closer to L as values of x are taken from *either* side of $x = a$. However, some functions exhibit different behaviors on the two sides of an x -value a , in which case it is necessary to distinguish whether values of x near a are on the left side or on the right side of a for purposes of investigating limiting behavior. For example, consider the function

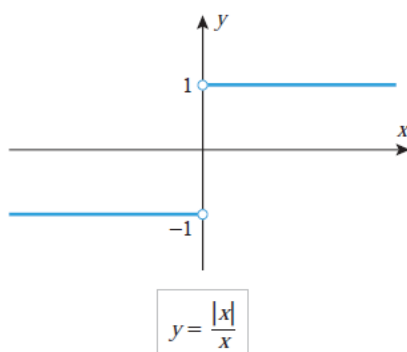
$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases} \quad (12)$$

which is graphed in Figure 1.1.12. As x approaches 0 from the *right*, the values of $f(x)$ approach a limit of 1 [in fact, the values of $f(x)$ are exactly 1 for all such x], and similarly, as x approaches 0 from the *left*, the values of $f(x)$ approach a limit of -1 . We denote these limits by writing

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1 \quad (13)$$

With this notation, the superscript “+” indicates a limit from the right and the superscript “−” indicates a limit from the left.

This leads to the general idea of a *one-sided limit*.



1.1.2 ONE-SIDED LIMITS (AN INFORMAL VIEW) If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but greater than a), then we write

$$\lim_{x \rightarrow a^+} f(x) = L \quad (14)$$

and if the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but less than a), then we write

$$\lim_{x \rightarrow a^-} f(x) = L \quad (15)$$

Expression (14) is read “the limit of $f(x)$ as x approaches a from the right is L ” or “ $f(x)$ approaches L as x approaches a from the right.” Similarly, expression (15) is read “the limit of $f(x)$ as x approaches a from the left is L ” or “ $f(x)$ approaches L as x approaches a from the left.”

THE RELATIONSHIP BETWEEN ONE-SIDED LIMITS AND TWO-SIDED LIMITS

In general, there is no guarantee that a function f will have a two-sided limit at a given point a ; that is, the values of $f(x)$ may not get closer and closer to any *single* real number L as $x \rightarrow a$. In this case we say that

$$\lim_{x \rightarrow a} f(x) \text{ does not exist}$$

Similarly, the values of $f(x)$ may not get closer and closer to a single real number L as $x \rightarrow a^+$ or as $x \rightarrow a^-$. In these cases we say that

$$\lim_{x \rightarrow a^+} f(x) \text{ does not exist}$$

or that

$$\lim_{x \rightarrow a^-} f(x) \text{ does not exist}$$

In order for the two-sided limit of a function $f(x)$ to exist at a point a , the values of $f(x)$ must approach some real number L as x approaches a , and this number must be the same regardless of whether x approaches a from the left or the right. This suggests the following result, which we state without formal proof.

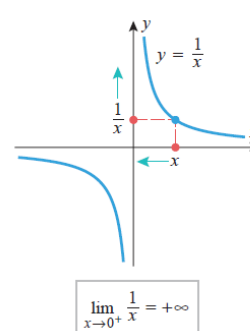
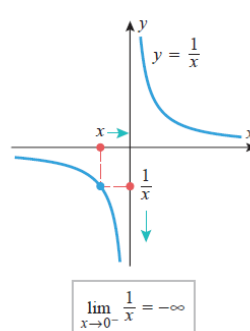
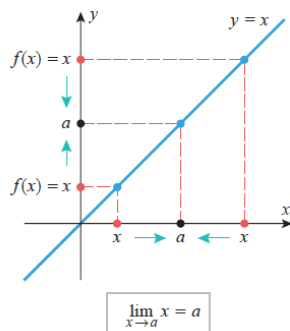
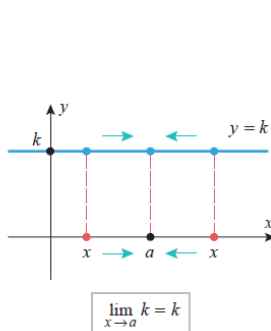
1.1.3 THE RELATIONSHIP BETWEEN ONE-SIDED AND TWO-SIDED LIMITS The two-sided limit of a function $f(x)$ exists at a if and only if both of the one-sided limits exist at a and have the same value; that is,

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

Computing Limits:

1.2.1 THEOREM Let a and k be real numbers.

$$(a) \lim_{x \rightarrow a} k = k \quad (b) \lim_{x \rightarrow a} x = a \quad (c) \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad (d) \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$



1.2.2 THEOREM Let a be a real number, and suppose that

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

That is, the limits exist and have values L_1 and L_2 , respectively. Then:

$$(a) \quad \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$$

$$(b) \quad \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$$

$$(c) \quad \lim_{x \rightarrow a} [f(x)g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right) = L_1 L_2$$

$$(d) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2}, \quad \text{provided } L_2 \neq 0$$

$$(e) \quad \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L_1}, \quad \text{provided } L_1 > 0 \text{ if } n \text{ is even.}$$

Moreover, these statements are also true for the one-sided limits as $x \rightarrow a^-$ or as $x \rightarrow a^+$.

1.2.3 THEOREM For any polynomial

$$p(x) = c_0 + c_1x + \cdots + c_nx^n$$

and any real number a ,

$$\lim_{x \rightarrow a} p(x) = c_0 + c_1a + \cdots + c_na^n = p(a)$$

Example: Find

$$(a) \lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} \quad (b) \lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12} \quad (c) \lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25}$$

Solution (a). The numerator and the denominator both have a zero at $x = 3$, so there is a common factor of $x - 3$. Then

$$\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)^2}{x - 3} = \lim_{x \rightarrow 3} (x - 3) = 0$$

Solution (b). The numerator and the denominator both have a zero at $x = -4$, so there is a common factor of $x - (-4) = x + 4$. Then

$$\lim_{x \rightarrow -4} \frac{2x + 8}{x^2 + x - 12} = \lim_{x \rightarrow -4} \frac{2(x + 4)}{(x + 4)(x - 3)} = \lim_{x \rightarrow -4} \frac{2}{x - 3} = -\frac{2}{7}$$

Solution (c). The numerator and the denominator both have a zero at $x = 5$, so there is a common factor of $x - 5$. Then

$$\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5} \frac{(x - 5)(x + 2)}{(x - 5)(x - 5)} = \lim_{x \rightarrow 5} \frac{x + 2}{x - 5}$$

However,

$$\lim_{x \rightarrow 5} (x + 2) = 7 \neq 0 \quad \text{and} \quad \lim_{x \rightarrow 5} (x - 5) = 0$$

so

$$\lim_{x \rightarrow 5} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5} \frac{x + 2}{x - 5}$$

does not exist. More precisely, the sign analysis in Figure 1.2.4 implies that

$$\lim_{x \rightarrow 5^+} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5^+} \frac{x + 2}{x - 5} = +\infty$$

and

$$\lim_{x \rightarrow 5^-} \frac{x^2 - 3x - 10}{x^2 - 10x + 25} = \lim_{x \rightarrow 5^-} \frac{x + 2}{x - 5} = -\infty \quad \blacktriangleleft$$

Limits Involving Radicals

Find $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$.

$$\frac{x - 1}{\sqrt{x} - 1} = \frac{(x - 1)(\sqrt{x} + 1)}{(\sqrt{x} - 1)(\sqrt{x} + 1)} = \frac{(x - 1)(\sqrt{x} + 1)}{x - 1} = \sqrt{x} + 1 \quad (x \neq 1)$$

Therefore,

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = \lim_{x \rightarrow 1} (\sqrt{x}+1) = 2 \quad \blacktriangleleft$$

Limits at Infinity

1.3.1 LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ eventually get as close as we like to a number L as x increases without bound, then we write

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow +\infty \quad (3)$$

Similarly, if the values of $f(x)$ eventually get as close as we like to a number L as x decreases without bound, then we write

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow -\infty \quad (4)$$

■ LIMIT LAWS FOR LIMITS AT INFINITY

It can be shown that the limit laws in Theorem 1.2.2 carry over without change to limits at $+\infty$ and $-\infty$. Moreover, it follows by the same argument used in Section 1.2 that if n is a positive integer, then

$$\lim_{x \rightarrow +\infty} (f(x))^n = \left(\lim_{x \rightarrow +\infty} f(x) \right)^n \quad \lim_{x \rightarrow -\infty} (f(x))^n = \left(\lim_{x \rightarrow -\infty} f(x) \right)^n \quad (9-10)$$

provided the indicated limit of $f(x)$ exists. It also follows that constants can be moved through the limit symbols for limits at infinity:

$$\lim_{x \rightarrow +\infty} kf(x) = k \lim_{x \rightarrow +\infty} f(x) \quad \lim_{x \rightarrow -\infty} kf(x) = k \lim_{x \rightarrow -\infty} f(x) \quad (11-12)$$

provided the indicated limit of $f(x)$ exists.

Finally, if $f(x) = k$ is a constant function, then the values of f do not change as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, so

$$\lim_{x \rightarrow +\infty} k = k \quad \lim_{x \rightarrow -\infty} k = k \quad (13-14)$$

1.3.2 INFINITE LIMITS AT INFINITY (AN INFORMAL VIEW) If the values of $f(x)$ increase without bound as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, then we write

$$\lim_{x \rightarrow +\infty} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = +\infty$$

as appropriate; and if the values of $f(x)$ decrease without bound as $x \rightarrow +\infty$ or as $x \rightarrow -\infty$, then we write

$$\lim_{x \rightarrow +\infty} f(x) = -\infty \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

as appropriate.

■ LIMITS OF x^n AS $x \rightarrow \pm\infty$

Figure 1.3.5 illustrates the end behavior of the polynomials x^n for $n = 1, 2, 3$, and 4. These are special cases of the following general results:

$$\lim_{x \rightarrow +\infty} x^n = +\infty, \quad n = 1, 2, 3, \dots$$

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} -\infty, & n = 1, 3, 5, \dots \\ +\infty, & n = 2, 4, 6, \dots \end{cases} \quad (15-16)$$

■ LIMITS OF RATIONAL FUNCTIONS AS $x \rightarrow \pm\infty$

One technique for determining the end behavior of a rational function is to divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, after which the limiting behavior can be determined using results we have already established. Here are some examples.

Example:

Find $\lim_{x \rightarrow +\infty} \frac{3x + 5}{6x - 8}$.

Solution. Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely, $x^1 = x$. We obtain

$$\begin{aligned}
 \lim_{x \rightarrow +\infty} \frac{3x + 5}{6x - 8} &= \lim_{x \rightarrow +\infty} \frac{3 + \frac{5}{x}}{6 - \frac{8}{x}} && \text{Divide each term by } x. \\
 &= \frac{\lim_{x \rightarrow +\infty} \left(3 + \frac{5}{x}\right)}{\lim_{x \rightarrow +\infty} \left(6 - \frac{8}{x}\right)} && \text{Limit of a quotient is the quotient of the limits.} \\
 &= \frac{\lim_{x \rightarrow +\infty} 3 + \lim_{x \rightarrow +\infty} \frac{5}{x}}{\lim_{x \rightarrow +\infty} 6 - \lim_{x \rightarrow +\infty} \frac{8}{x}} && \text{Limit of a sum is the sum of the limits.} \\
 &= \frac{3 + 5 \lim_{x \rightarrow +\infty} \frac{1}{x}}{6 - 8 \lim_{x \rightarrow +\infty} \frac{1}{x}} = \frac{3 + 0}{6 - 0} = \frac{1}{2} && \text{A constant can be moved through a limit symbol; Formulas (2) and (13).}
 \end{aligned}$$

Example:

Find

$$) \lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$$

Solution (b). Divide each term in the numerator and denominator by the highest power of x that occurs in the denominator, namely, $x^1 = x$. We obtain

$$\lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} = \lim_{x \rightarrow +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3} \quad (19)$$

In this case we cannot argue that the limit of the quotient is the quotient of the limits because the limit of the numerator does not exist. However, we have

$$\lim_{x \rightarrow +\infty} 5x^2 - 2x = +\infty, \quad \lim_{x \rightarrow +\infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow +\infty} \left(\frac{1}{x} - 3\right) = -3$$

Thus, the numerator on the right side of (19) approaches $+\infty$ and the denominator has a finite *negative* limit. We conclude from this that the quotient approaches $-\infty$; that is,

$$\lim_{x \rightarrow +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x} = \lim_{x \rightarrow +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3} = -\infty$$

Limits Involving Radicals

Example: Find

$$\lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3)$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3) &= \lim_{x \rightarrow +\infty} (\sqrt{x^6 + 5} - x^3) \left(\frac{\sqrt{x^6 + 5} + x^3}{\sqrt{x^6 + 5} + x^3} \right) \\&= \lim_{x \rightarrow +\infty} \frac{(x^6 + 5) - x^6}{\sqrt{x^6 + 5} + x^3} = \lim_{x \rightarrow +\infty} \frac{5}{\sqrt{x^6 + 5} + x^3} \\&= \lim_{x \rightarrow +\infty} \frac{\frac{5}{x^3}}{\sqrt{1 + \frac{5}{x^6}} + 1} \quad \boxed{\sqrt{x^6} = x^3 \text{ for } x > 0} \\&= \frac{0}{\sqrt{1 + 0} + 1} = 0\end{aligned}$$

■ LIMITS OF PIECEWISE-DEFINED FUNCTIONS

For functions that are defined piecewise, a two-sided limit at a point where the formula changes is best obtained by first finding the one-sided limits at that point.

► **Example 11** Let

$$f(x) = \begin{cases} 1/(x+2), & x < -2 \\ x^2 - 5, & -2 < x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$$

Find

$$(a) \lim_{x \rightarrow -2^-} f(x) \quad (b) \lim_{x \rightarrow 0} f(x) \quad (c) \lim_{x \rightarrow 3} f(x)$$

Solution (a). We will determine the stated two-sided limit by first considering the corresponding one-sided limits. For each one-sided limit, we must use that part of the formula that is applicable on the interval over which x varies. For example, as x approaches -2 from the left, the applicable part of the formula is

$$f(x) = \frac{1}{x+2}$$

and as x approaches -2 from the right, the applicable part of the formula near -2 is

$$f(x) = x^2 - 5$$

Thus,

$$\begin{aligned} \lim_{x \rightarrow -2^-} f(x) &= \lim_{x \rightarrow -2^-} \frac{1}{x+2} = -\infty \\ \lim_{x \rightarrow -2^+} f(x) &= \lim_{x \rightarrow -2^+} (x^2 - 5) = (-2)^2 - 5 = -1 \end{aligned}$$

from which it follows that $\lim_{x \rightarrow -2} f(x)$ does not exist.

Solution (b). The applicable part of the formula is $f(x) = x^2 - 5$ on both sides of 0, so there is no need to consider one-sided limits here. We see directly that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x^2 - 5) = 0^2 - 5 = -5$$

Solution (c). Using the applicable parts of the formula for $f(x)$, we obtain

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} (x^2 - 5) = 3^2 - 5 = 4 \\ \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} \sqrt{x+13} = \sqrt{\lim_{x \rightarrow 3^+} (x+13)} = \sqrt{3+13} = 4 \end{aligned}$$

Since the one-sided limits are equal, we have

$$\lim_{x \rightarrow 3} f(x) = 4$$

Indeterminate Forms:

Indeterminate forms of $0/0$ and ∞/∞

3.6.1 THEOREM (L'Hôpital's Rule for Form $0/0$) Suppose that f and g are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

If $\lim_{x \rightarrow a} [f'(x)/g'(x)]$ exists, or if this limit is $+\infty$ or $-\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Moreover, this statement is also true in the case of a limit as $x \rightarrow a^-$, $x \rightarrow a^+$, $x \rightarrow -\infty$, or as $x \rightarrow +\infty$.

3.6.2 THEOREM (L'Hôpital's Rule for Form ∞/∞) Suppose that f and g are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that

$$\lim_{x \rightarrow a} f(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \infty$$

If $\lim_{x \rightarrow a} [f'(x)/g'(x)]$ exists, or if this limit is $+\infty$ or $-\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Moreover, this statement is also true in the case of a limit as $x \rightarrow a^-$, $x \rightarrow a^+$, $x \rightarrow -\infty$, or as $x \rightarrow +\infty$.

► **Example 1** Find the limit

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

using L'Hôpital's rule, and check the result by factoring.

Solution. The numerator and denominator have a limit of 0, so the limit is an indeterminate form of type $0/0$. Applying L'Hôpital's rule yields

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{d}{dx}[x^2 - 4]}{\frac{d}{dx}[x - 2]} = \lim_{x \rightarrow 2} \frac{2x}{1} = 4$$

This agrees with the computation

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4 \quad \blacktriangleleft$$

► **Example 5** Evaluate $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$.

Solution. Both terms have a limit of $+\infty$, so the stated problem is an indeterminate form of type $\infty - \infty$. Combining the two terms yields

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \sin x}$$

which is an indeterminate form of type $0/0$. Applying L'Hôpital's rule twice yields

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \sin x} &= \lim_{x \rightarrow 0^+} \frac{\cos x - 1}{\sin x + x \cos x} \\ &= \lim_{x \rightarrow 0^+} \frac{-\sin x}{\cos x + \cos x - x \sin x} = \frac{0}{2} = 0 \quad \blacktriangleleft \end{aligned}$$

Example:

Find $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$.

Solution. As discussed above, we begin by introducing a dependent variable

$$y = (1 + \sin x)^{1/x}$$

and taking the natural logarithm of both sides:

$$\ln y = \ln(1 + \sin x)^{1/x} = \frac{1}{x} \ln(1 + \sin x) = \frac{\ln(1 + \sin x)}{x}$$

Thus,

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$$

which is an indeterminate form of type $0/0$, so by L'Hôpital's rule

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x} = \lim_{x \rightarrow 0} \frac{(\cos x)/(1 + \sin x)}{1} = 1$$

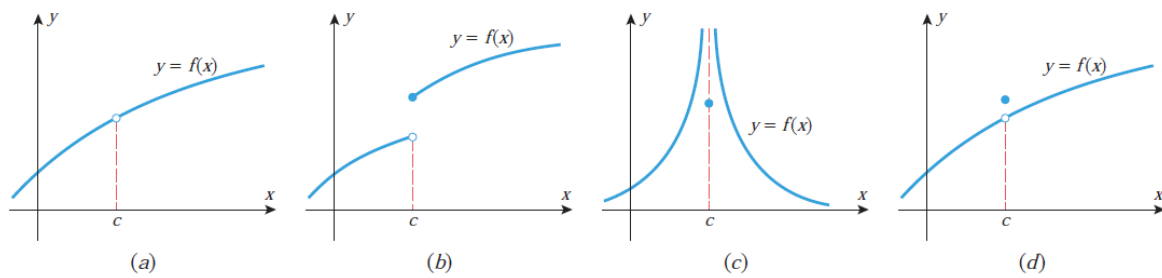
Since we have shown that $\ln y \rightarrow 1$ as $x \rightarrow 0$, the continuity of the exponential function implies that $e^{\ln y} \rightarrow e^1$ as $x \rightarrow 0$, and this implies that $y \rightarrow e$ as $x \rightarrow 0$. Thus,

$$\lim_{x \rightarrow 0} (1 + \sin x)^{1/x} = e \quad \blacktriangleleft$$

Continuity:

1.5.1 DEFINITION A function f is said to be *continuous at $x = c$* provided the following conditions are satisfied:

1. $f(c)$ is defined.
2. $\lim_{x \rightarrow c} f(x)$ exists.
3. $\lim_{x \rightarrow c} f(x) = f(c)$.



▲ Figure 1.5.1

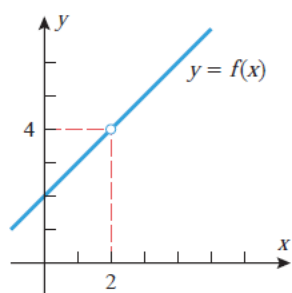
► **Example 1** Determine whether the following functions are continuous at $x = 2$.

$$f(x) = \frac{x^2 - 4}{x - 2}, \quad g(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 3, & x = 2, \end{cases} \quad h(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$$

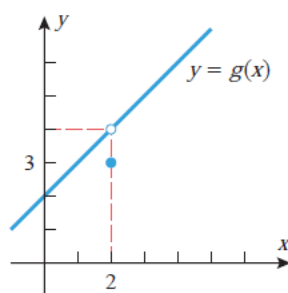
Solution. In each case we must determine whether the limit of the function as $x \rightarrow 2$ is the same as the value of the function at $x = 2$. In all three cases the functions are identical, except at $x = 2$, and hence all three have the same limit at $x = 2$, namely,

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$$

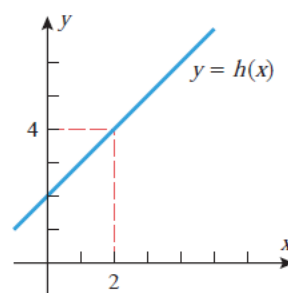
The function f is undefined at $x = 2$, and hence is not continuous at $x = 2$ (Figure 1.5.2a). The function g is defined at $x = 2$, but its value there is $g(2) = 3$, which is not the same as the limit as x approaches 2; hence, g is also not continuous at $x = 2$ (Figure 1.5.2b). The value of the function h at $x = 2$ is $h(2) = 4$, which is the same as the limit as x approaches 2; hence, h is continuous at $x = 2$ (Figure 1.5.2c). (Note that the function h could have been written more simply as $h(x) = x + 2$, but we wrote it in piecewise form to emphasize its relationship to f and g .) ◀



(a)



(b)



(c)